

Geomodelling - Geostatistics for Natural Resource Evaluation

Report Explorative Data Analysis (EDA)

I) Task description

The aim of this report is to do an **Exploratory Data analysis** for the spatial dataset testData123.csv. In this analysis I look at the structure of the data, the position of the sample points, the distribution of the measured values, possible Outliers and also possible correlation between the variables. The goal is to find if there are unusual things in the data, for example skewed data, rounded values, spatial trends, jumps or maybe faults in the map.

The analysis is done with **R** and the important plots and statistical results are used to explain the findings.

II) Description of the dataset

- The dataset contains five variables: PointID, x, y, Co and Ni.
- **PointID** is the identification number of each sampling point.
- The variables x and y describe the spatial coordinates of the samples.
- **Co** and **Ni** are numeric measured variables.
- The coordinates and PointID are stored as **integer** variables, while Co and Ni are stored as **numeric** variables.

Quantities:

Number of rows and columns : **(994, 5)**

Variables: **"PointID"** **"x"** **"y"** **"Co"** **"Ni"**

Data types:

PointID => **"Integer"**

x => **"integer"**

y => **"integer"**

Co => **"numeric"**

Ni => **"numeric"**

Missing values: **No Missing value**

min, max, mean, median and quartile values:

PointID	x	y	Co	Ni
Min. : 1.0	Min. : 0	Min. : 0	Min. : 0.766	Min. : 2.731
1st Qu.: 249.2	1st Qu.: 800	1st Qu.: 800	1st Qu.: 7.554	1st Qu.:13.978
Median : 497.5	Median :1500	Median :1500	Median :10.236	Median :17.022
Mean : 554.7	Mean :1527	Mean :1508	Mean :10.655	Mean :18.038
3rd Qu.: 745.8	3rd Qu.:2300	3rd Qu.:2300	3rd Qu.:13.042	3rd Qu.:21.820
Max. :8610.0	Max. :3271	Max. :3000	Max. :24.368	Max. :36.554

Variance of Co and Ni:

var(Co) = **18.26951**

var(Ni) = **33.83242**

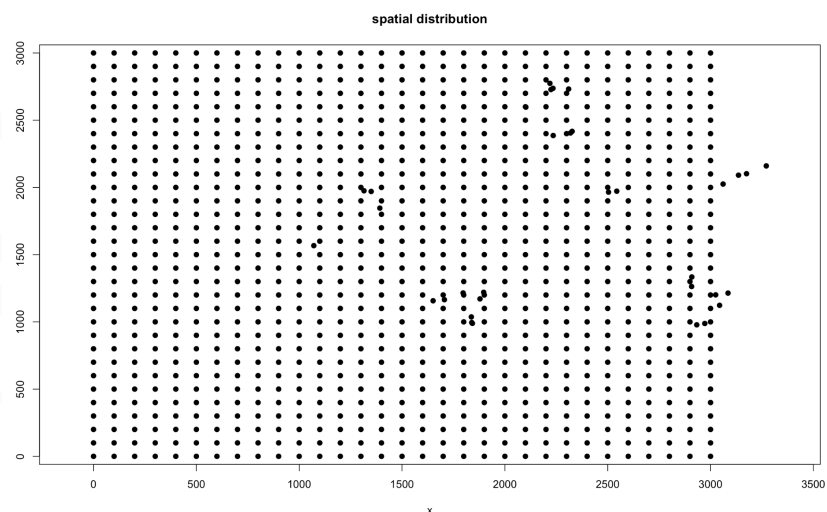
Note: The variables x and y are the spatial coordinates of the sampling points.

Note: The minimum and maximum x-coordinates are 0 and 3271, and the minimum and maximum y-coordinates are 0 and 3000.

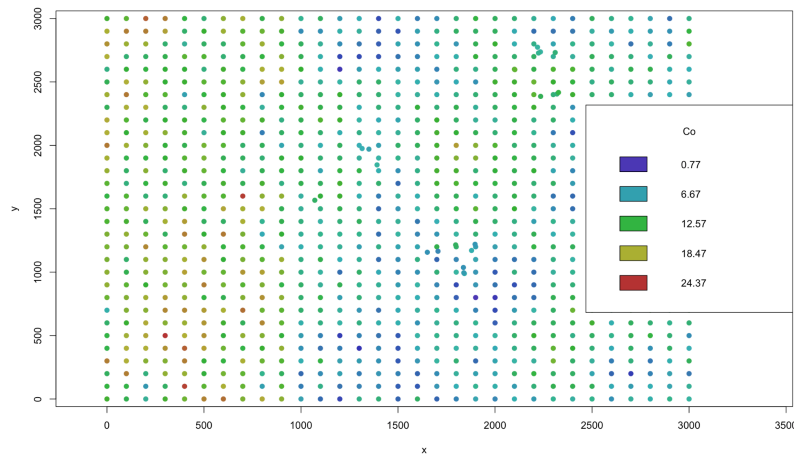
Note: For Co, the values range from 0.766 to 24.368. The mean value is 10.655 and the median is 10.236.

Note: For Ni, the values range from 2.731 to 36.554. The mean value is 18.038 and the median is 17.022. The variance of Co is 18.26951 and the variance of Ni is 33.83242.

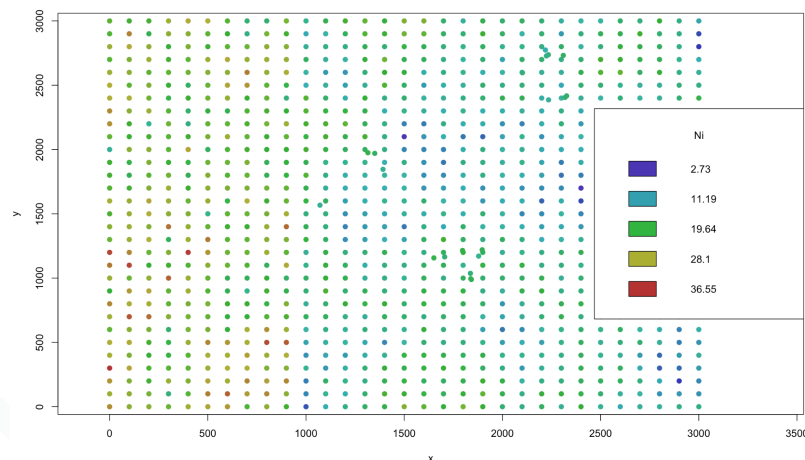
Spatial distribution:



Spatial distribution of the sampling points



Spatial distribution of Co values

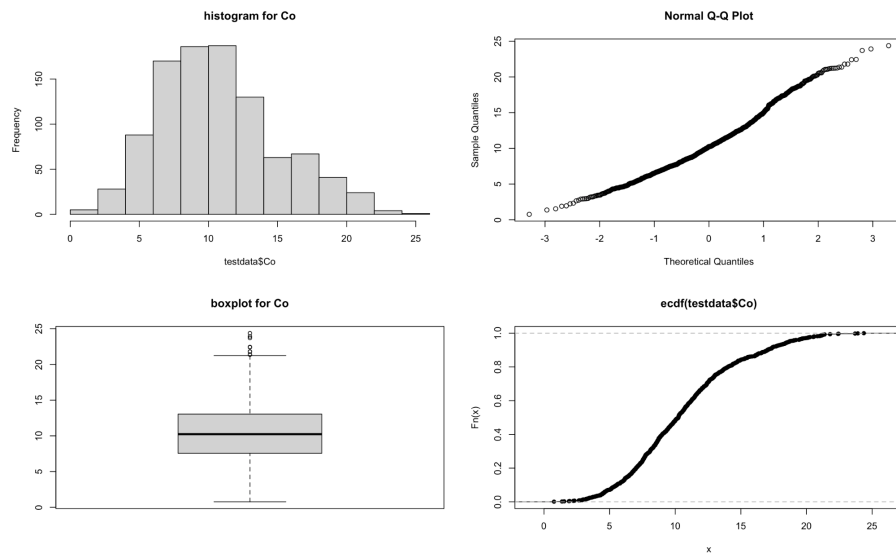


Spatial distribution of Ni values

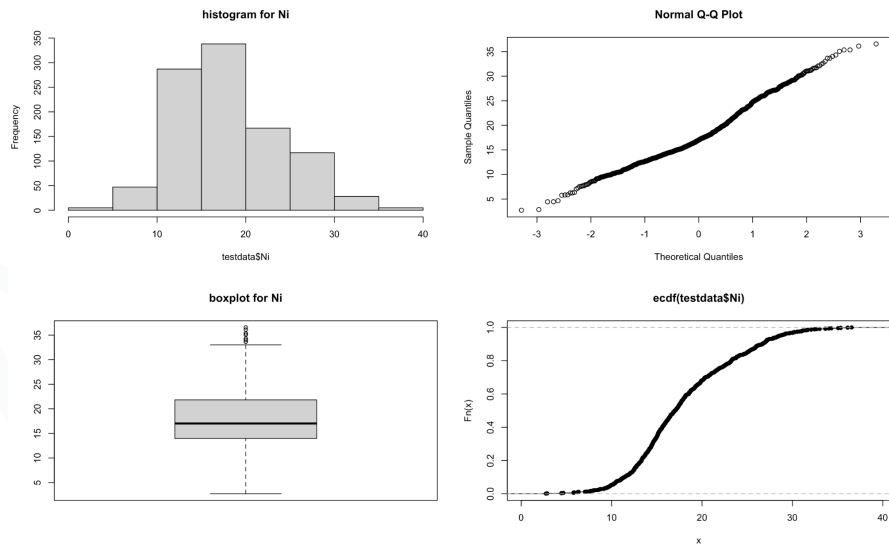
The sampling locations are mostly arranged in a regular grid. The spacing between most points seems to be constant. Some points are not exactly on the regular grid and a few points are located outside the main frame, especially on the right side of the study area. No large empty holes can be seen inside the main sampling area, but there are some local irregularities and small clusters.

The first spatial plots of Co and Ni show that the measured values have a spatial structure. Higher values of Co and Ni occur more often in the western and south-western part of the frame. Lower values are more visible in the central and eastern parts. This first visual impression suggests that Co and Ni may have a similar spatial pattern.

III) Distribution



Histogram, qq-plot, boxplot and empirical distribution function of **Co**



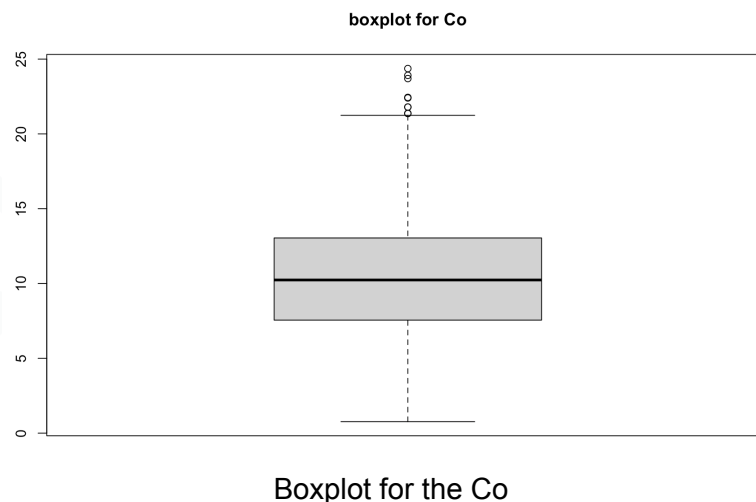
Histogram, qq-plot, boxplot and empirical distribution function of **Ni**

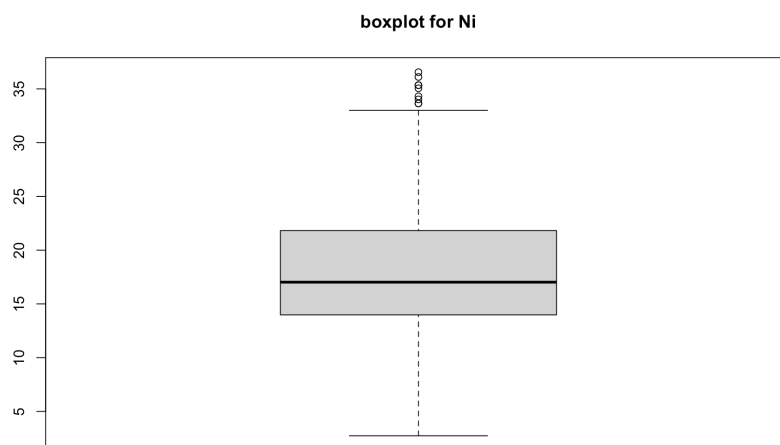
For Co, the values range from 0.766 to 24.368. The mean value is 10.655 and the median is 10.236. Because the mean is slightly higher than the median, the distribution seems to be slightly **right-skewed**. The histogram also shows one main peak around the middle values and a longer tail to the right side. The QQ-plot does not follow a straight line, especially at the lower and upper ends, so the Co values are **not normally distributed**.

For Ni, the values range from 2.731 to 36.554. The mean value is 18.038 and the median is 17.022. The mean is also higher than the median, which indicates a **right-skewed** distribution. The histogram shows that most values are in the lower and middle range, while fewer values occur at the high end. The QQ-plot also shows deviations from normality, especially in the tails. Therefore, Ni is also **not normally distributed**.

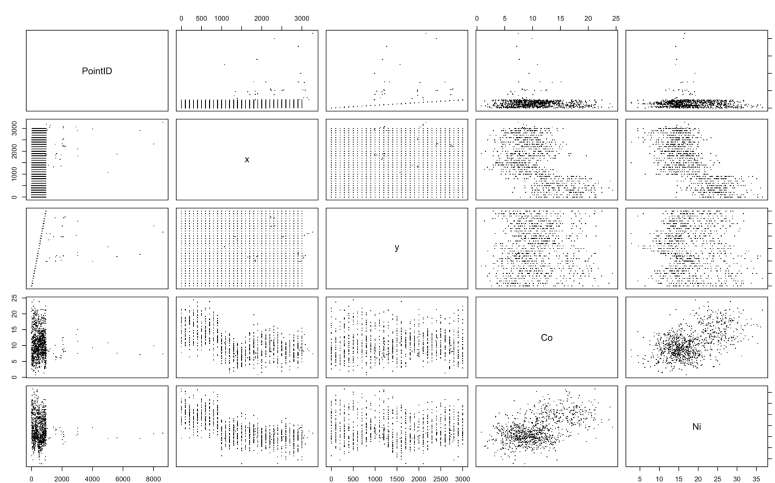
IV) Outliers

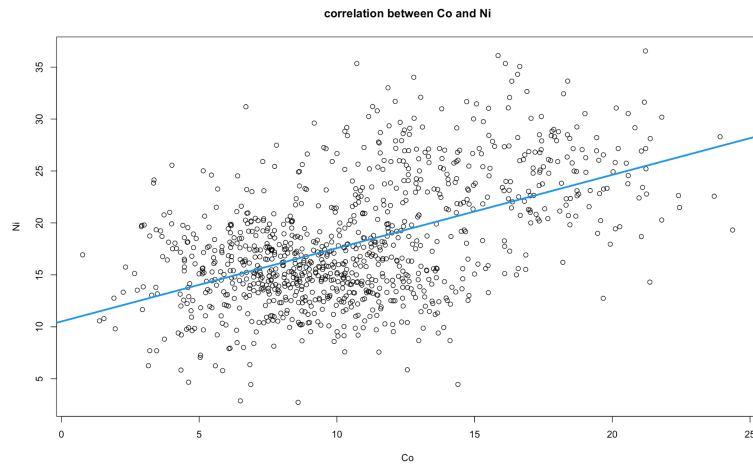
For Co, the boxplot shows some high outliers above the upper part. These values are around 22 to 24. There are no clear lower outliers. This means that a small number of samples have much higher Co values than most other samples. For Ni, the boxplot also shows some high outliers above the upper part. These values are around 34 to 37. There are no clear lower outliers. Therefore, both Co and Ni have high-value outliers. These outliers cause the right-skewed distribution seen in the histograms.





V) Correlation





scatter plot and linear regression line between co and ni

The correlation coefficient between Co and Ni is **0.519**. This shows a moderate positive correlation. It means that higher Co values are often connected with higher Ni values, but the relationship is **not very strong** because the points are still quite spread.

The scatter plot also shows an increasing trend between Co and Ni. The fitted regression line has the equation **$Ni = 10.5169 + 0.7059 \times Co$** . This means that if Co increases by one unit, Ni increases on average by about 0.706 units.

The correlation matrix also shows negative correlations between the x-coordinate and the two measured variables. The correlation between x and co is -0.537, and the correlation between x and Ni is -0.680. This can indicate a spatial trend, where co and Ni values tend to decrease from west to east.

VI) Conclusions

In this exploratory data analysis, the dataset **testData123.csv** was analysed using **summary statistics, spatial plots, histograms, QQ-plots, boxplots, empirical distribution functions and correlation analysis**.

The dataset contains **994** sampling points with five variables: **PointID, x, y, Co** and **Ni**. The sampling points are mostly arranged in a regular grid structure. Some points are slightly outside the main grid and there are some local irregularities, but no large empty holes can be seen inside the main sampling area.

The spatial plots show that Co and Ni are **not randomly distributed**. Higher values occur more often in the western and south-western part of the study area, while lower values are more visible in the central and eastern parts. This indicates a **spatial trend** in the data.

The distributions of Co and Ni are not perfectly normal. Both variables are slightly **right-skewed**, because the **mean values are higher than the median values** and the histograms have longer right tails. The QQ-plots also show deviations from a normal distribution, especially in the tails.

The boxplots show high-value outliers for both Co and Ni. For Co, the outliers are around 22 to 24. For Ni, the outliers are around 34 to 37. **These outliers are probably one reason for the right-skewed distributions.**

The correlation analysis shows a **moderate positive correlation** between Co and Ni. The correlation coefficient is **0.519**. This means that higher Co values are often connected with higher Ni values, but the relation is **not very strong**. The regression line also shows an increasing linear trend. In addition, both Co and Ni have **negative correlations with the x-coordinate**, which supports the idea of a west-east spatial trend.

Overall, the dataset shows some important spatial structure, right-skewed distributions, high-value outliers and a moderate positive dependency between Co and Ni. No clear fault or sharp jump can be proven only from these plots, but some spatial changes and trends are visible in the maps.